

Stellar structure of quark stars in a modified Starobinsky gravity

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Introduction

Extended theories of General Relativity

- The failure of GR at high curvature scenarios such as black hole singularity and low curvature dilemma such as late-time cosmic acceleration, flat rotation curves etc tell us of the inadequacy of the theory¹.
- Starobinsky proposed a non-linear curvature theory generated by considering coupling between classical gravitational field and quantum matter fields², generating
 - ▶ $\langle T_{\mu\nu} \rangle = c_1(R_{\mu}^{\alpha} R_{\nu\alpha} - \frac{2}{3}RR_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R_{\alpha\beta}R^{\alpha\beta} + \frac{1}{4}g_{\mu\nu}R^2) + \frac{c_2}{6}(2R_{;\mu;\nu} - 2g_{\mu\nu}\square R - 2RR_{\mu\nu} + \frac{1}{2}g_{\mu\nu}R^2)$
 - ▶ He showed the existence of one-parameter family of non-singular solution of de-Sitter type.
 - ▶ This de-Sitter phase naturally explains inflation without invoking an inflaton field.
- The above terms can be generated from a Lagrangian of polynomial function of R .
- This leads to the proposal of a generalized Einstein Hilbert action of the form $S = \int \sqrt{-g}f(R)d^4x$.
- It was shown that a quadratic form $R + \alpha R^2$ could explain inflation with better fit to the observational data³.
- It explains the late time cosmic acceleration⁴.
- Both flat rotation curves and MOND can be explain from simple power-law form⁵ of $f(R)$.

¹ A. A. Starobinskii, Sov. Astron. Lett. 9, 302 (1983), A. G. Riess et al, Astron. J 116(3), 1009 (1998), J. H. Oort, Bull. Astron. Inst. (Netherlands) 6, 249 (1932)

² A. Starobinsky, Phys. Lett. B 91, 99 (1980)

³ S. M. Carroll et al., Phys. Rev. D 70, 043528 (2004), M. Artymowski et al., J. Cosmol. Astropart. Phys. 09, 036 (2014)

⁴ S. Capozziello et al., Phys. Lett. A 326, 292 (2004), S. Capozziello et al., Mon. Not. Roy. Astron. Soc. 375, 1423 (2007)

⁵ S. Mendoza, Can. J. Phys. 93, 217 (2015)

Introduction to

$f(R, T)$ gravity (where T is trace of stress-energy tensor)

- $f(R, T)$ was first proposed by T. Harko et al.⁶ in 2011. He speculated that the dependence on T may be induced by exotic imperfect fluids or quantum effects.
- It is shown that the action of the form $S = \frac{1}{16\pi} \int f(R, T) \sqrt{-g} d^4x + \int L_m \sqrt{-g} d^4x$ gives the field equation as

$$f_R R_{\mu\nu} - \frac{1}{2} f g_{\mu\nu} + (g_{\mu\nu} \nabla^\alpha \nabla_\alpha - \nabla_\mu \nabla_\nu) f_R = \kappa T_{\mu\nu} - f_T T_{\mu\nu} - f_T \Theta_{\mu\nu}$$

where $\Theta_{\mu\nu} = g^{\alpha\beta} \delta T_{\alpha\beta} / \delta g^{\mu\nu}$, $f_R = \partial f / \partial R$ and $f_T = \partial f / \partial T$.

- Covariant divergence of the above field equation gives

$$\nabla^\mu T_{\mu\nu} = \frac{f_T}{\kappa + f_T} \left[(p g_{\mu\nu} - T_{\mu\nu}) \nabla^\mu \ln f_T + g_{\mu\nu} \nabla^\mu p \right] \quad (1)$$

Note that we could recover Einstein equation for $f(R, T) = R$.

- It is interesting to note that the above equation $\nabla^\mu T_{\mu\nu} \neq 0$ leads to a non-geodesic equation, for perfect fluid

$$\frac{d^2 x^\mu}{ds^2} + \Gamma_{\nu\lambda}^\mu u^\nu u^\lambda = 8\pi \frac{\nabla_\nu p}{(\varepsilon + p)[8\pi + f_T]} (g^{\mu\nu} - u^\mu u^\nu) \quad (2)$$

⁶T. Harko, F. S. N. Lobo, S. Nojiri, S.D. Odintsov, Phys. Rev. D 84, 024020 (2011)

Multiplicative form of $f(R, T)$ gravity

$$f(R, T) = (R + \alpha R^2)(1 + \omega T)$$

- We considered $f(R, T)$ to take account of possible quantum effects in addition to the Starobinsky gravity.
- To the first order in ω , the field equations are obtained as

$$f_R G_{\mu\nu} - \frac{1}{2}(f - f_R R)g_{\mu\nu} + (g_{\mu\nu}\square - \nabla_\mu \nabla_\nu)f_R = \kappa T_{\mu\nu} + \omega \left[(f_0 - kT_0)T_{0\mu\nu} - f_{R_0}(g_{0\mu\nu}\square - \nabla_\mu \nabla_\nu)T_0 - f_0 p_0 g_{0\mu\nu} - \left(2g_{0\mu\nu}g_0^{\alpha\beta} - \delta_\mu^\alpha \delta_\nu^\beta - \delta_\mu^\beta \delta_\nu^\alpha \right) \frac{\partial f_{R_0}}{\partial x^\alpha} \frac{\partial T_0}{\partial x^\beta} \right]$$

where the subscript “0” indicates unperturbed quantities when $\omega = 0$.

- And the corresponding trace becomes

$$3\square f_R + f_R R - 2f = \kappa T + \omega \left[(f_0 - kT_0)T_0 - 3f_{R_0}\square T_0 - \left(8g_0^{\alpha\beta} - 2g_0^{\mu\nu}\delta_\mu^\alpha \delta_\nu^\beta \right) \frac{\partial f_{R_0}}{\partial x^\alpha} \frac{\partial T_0}{\partial x^\beta} - 4f_0 p_0 \right]$$

- For $\omega = 0$, we recover the corresponding equation in $f(R)$ gravity.

Field equations for spherically symmetric stars

The trace equation is

$$R'' = -\frac{2}{r}R' + \frac{\kappa}{6\alpha}Te^\lambda + \frac{e^\lambda}{6\alpha}R + \frac{\lambda' - \nu'}{2}R' + \frac{\omega}{6\alpha} \left[f_0 - \kappa T_0 \right] T_0 e^{\lambda_0} - 6f'_{R_0} T'_0 - 3\phi_{R_0} T''_0 \\ - 4f_0 p_0 e^{\lambda_0} - 3f_{R_0} \left(\frac{\nu'_0 - \lambda'_0}{2} + \frac{2}{r} \right) T'_0 \Big],$$

The tt -component of the field equation is

$$\lambda' = \frac{1 - e^\lambda}{r} + \frac{1}{6} \frac{re^\lambda}{\phi_R} (2 + 3\alpha R) R - \gamma \nu' + \frac{\kappa}{3} \frac{re^\lambda}{\phi_R} (3\varepsilon + T) + \omega \left[\frac{re^{\lambda_0}}{3\phi_{R_0}} \{ (\phi(R_0) - \kappa T_0)(3\varepsilon_0 + T_0) - \phi(R_0)p_0 \} \right. \\ \left. - rT'_0 \frac{\nu'_0}{2} \right], \quad (3)$$

The rr -component of the field equation is

$$\nu' = \frac{1}{1 + \gamma} \left[\frac{\kappa re^\lambda}{\phi_R} p + \frac{e^\lambda - 1}{r} - \frac{1}{2} \frac{re^\lambda}{\phi_R} \alpha R^2 - \frac{4}{r} \gamma \right] - \frac{\omega}{1 + \gamma_0} \left[\frac{re^{\lambda_0}}{\phi_{R_0}} \kappa T_0 p_0 + r \left(\frac{\nu'_0}{2} + \frac{2}{r} \right) T'_0 \right]$$

and the covariant derivative of the field equation leads to

$$\rho' = -(\varepsilon + p) \frac{\nu'}{2} - \omega(\varepsilon_0 + p_0) \frac{\phi(R_0)}{\kappa} \frac{\nu'_0}{2} \quad (4)$$

where R_0 , λ_0 , ν_0 , ε_0 and p_0 are the solutions of the above field equations when ω is set equal to zero (the pure Starobinsky case).

Far Field solution

- In the region exterior to the star, the trace equation is

$$3\Box f_R + f_R R - 2f = 0 \quad (5)$$

which is identical to that obtained for $f(R)$ gravity in vacuum.

- For $f(R) = R + \alpha R^2$, the above equation reduces to

$$6\Box R - R = 0. \quad (6)$$

- For spherically symmetric static metric this equation becomes

$$e^{-\lambda} \left\{ R'' + \left(\frac{\nu'}{2} - \frac{\lambda'}{2} + \frac{2}{r} \right) R' \right\} - \frac{R}{6\alpha} = 0 \quad (7)$$

- The Starobinsky correction αR^2 is a very weak contribution as one approaches infinity. The last term dominates in the limit $\alpha \rightarrow 0$ giving us back $R = 0$ or the Schwarzschild solution.
- Thus ν and λ and their first derivatives are expected to approach zero on approaching infinity.
- At far field we have

$$R'' + \frac{2}{r} R' - \frac{R}{6\alpha} = 0 \quad (8)$$

whose solution is simply given by

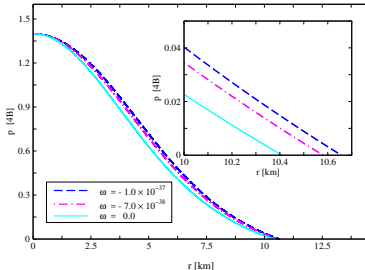
$$R(r) = c_1 \frac{e^{-\frac{r}{\sqrt{6\alpha}}}}{r}, \quad (9)$$

- For $\alpha > 0$ the Ricci scalar decays as Yukawian which approaches zero faster than r^{-1} as $r \rightarrow \infty$ whereas for $\alpha < 0$, it is oscillatory.

Interior and exterior solutions

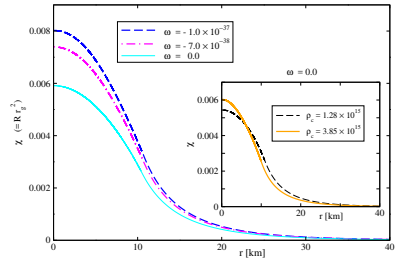
Pressure profile and Ricci scalar profile

- The pressure profile for different ω with same central density ρ_c is shown.
- For $\omega \neq 0$, we see an increase in the pressure gradient slightly away from the center.



- The additional term proportional to ω reduces the gravitational force.
- This dilutes the density if the star and pushes the stellar boundary away as shown in the inset.

- Radial profile of scalar curvature $R(r)$ for different ω with same ρ_c .
- The terms proportional to ω is similar to the term $\frac{\kappa}{6\alpha} T e^\lambda$.

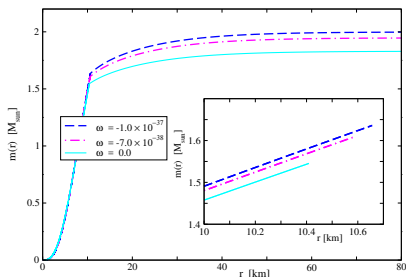


- The terms proportional to ω effectively changes the matter content (T) with respect to the quadratic gravity.
- As ω is increased the matter content is increased.
- Thus for larger ω value, R takes higher value at the center.

Interior and exterior solutions

Mass profile

- The cumulative mass profile is shown for different ω values for constant central density.
- With respect to the quadratic gravity there is slight increase in the stellar mass M_s as ω is increased.

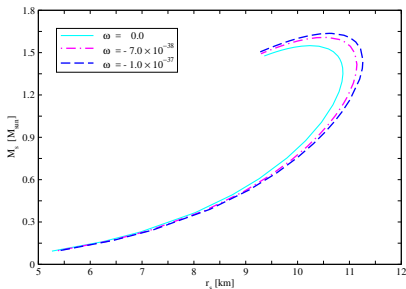


- But there is an appreciable increase in the total mass M seen by the distant observer as the ω value is increased.

- Outside the stellar radius the mass still increases unlike Einstein gravity.
- As we have seen earlier, the Ricci scalar is non-vanishing outside the star, this non-zero value of scalar curvature contribute significantly to the total mass M .
- The matter content of the star dictates the central R_c value, the magnitude of R_c in turn determines the difference $\Delta M = M - M_s$.
- The radius of the gravitational halo r_H similar to the stellar radius r_s are negligibly affected by the contribution from the $\omega \neq 0$.
- Outside the gravitational halo r_H , Ricci scalar is $R \approx 0$, implying Schwarzschild solution.

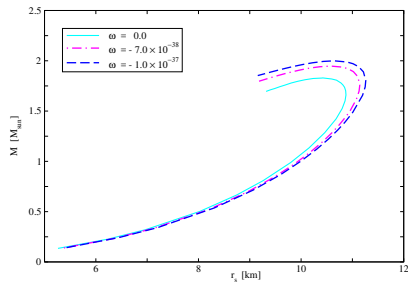
Mass-radius ($\mathcal{M} - \mathcal{R}$) relation

- Mass-radius relation between stellar mass M_s and stellar Radius r_s
- We see an increment in both, the maximal stellar mass and the corresponding stellar radius.



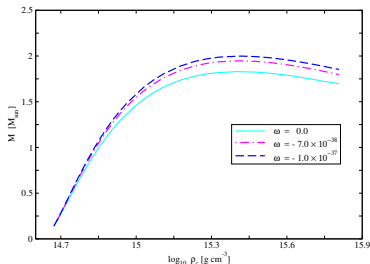
ω ($\text{cm}^3 \text{erg}^{-1}$)	ρ_c^* (g cm^{-3})	R_c^* ($\times 10^{-3} r_g^{-2}$)	r_s^* (km)	M_s^* ($M_{\odot}$)	M^* ($M_{\odot}$)
0.0	2.5412×10^{15}	5.90834	10.4201	1.5447	1.8298
-7.0×10^{-38}	2.5669×10^{15}	7.39491	10.5807	1.6076	1.9472
-1.0×10^{-37}	2.6097×10^{15}	8.08182	10.6334	1.6360	1.9988

- Mass-radius relation between total mass M and stellar Radius r_s .
- We see an increment in maximal total mass M^* .



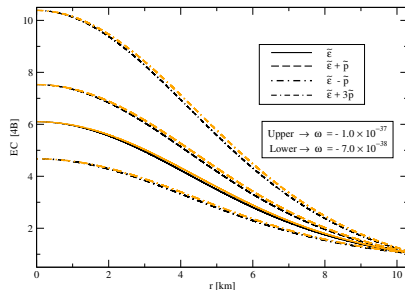
Stability and energy conditions

- Total mass M as a function of central density ρ_c is examined stability analysis.
- The region in the mass-central density curve, $\frac{\partial M}{\partial \rho_c} > 0$, corresponds to the stable configurations.
- The unstable region is given by $\frac{\partial M}{\partial \rho_c} < 0$.
- The onset of instability is identified as the point where $\partial M / \partial \rho_c = 0$, and the mass corresponding to this point is the maximal^a.



^aS. A. Teukolsky, Stuart L. Shapiro, Black holes, white dwarfs, and neutron stars, (Wiley, 1983)

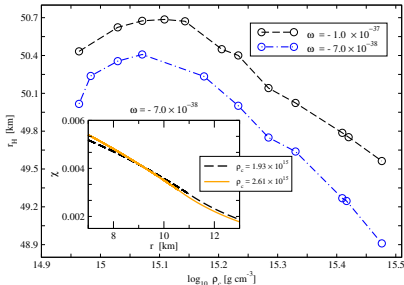
- Energy conditions are shown as a function of radial coordinate.



- Null energy condition ($\epsilon \geq 0$),
- Weak energy condition ($\epsilon + p \geq 0$)
- Strong energy condition ($\epsilon + 3p \geq 0$)
- Dominant energy condition ($\epsilon - |p| \geq 0$)
- We see that all energy conditions are satisfied and they are positive in the entire region of the star.

Gravitational Halo

- The effective radius of gravitational halo r_H as a function of central density is shown until the onset of gravitational instability.
- The radius r_H decreases with increasing central density ρ_c .



- The Schwarzschild radius $r_{\text{Sch}} = \frac{2GM}{c^2}$ increases with increasing central density ρ_c .

- Thus we see expansion of the Schwarzschild radius r_{Sch} and shrinkage of the gravitational halo r_H with increasing central density.
- When the star undergoes a gravitational collapse, it suggests that the gravitational halo will be pulled inside the Event Horizon.
- Thus only possible solution outside the Black Hole is Schwarzschild solution^a.
- Whereas $R \neq 0$ solution is still valid to the exterior of the star.

^aS. Mignemi et al., Phys. Rev. D 46, 1475 (1992)

Concluding Remarks

- Einstein gravity is successful in low gravity phenomena in the solar system.
- However Einstein gravity is inadequate to describe phenomena related to intense gravity (Neutron stars) as well as very weak gravity (such flat rotation curves in galaxy)
- In this work we took up the case of intense gravity in a quark star.
- In a quark stars for which a modified gravity such as $f(R)$ or $f(R, T)$ is required.
- $f(R)$ gravity has not be able to explain mass as high as $\sim 2 M_{\odot}$.
- The present theory based on $f(R, T)$ is capable of explaining mass as high as $\sim 2 M_{\odot}$ in the case of quark star.
- We thus guess that the observed high mass pulsar $\sim 2 M_{\odot}$ have a large quark core with a thin neutron crust⁷.

⁷ J. Antoniadis et al. Science 340 1233232 (2013), P. B. Demorest et al., Nature 467, 1081 (2010), E. Fonseca et al., Astrophys. J. 832, 167 (2016), H. T. Cromartie et al., Nat. Astron. (2019)

Future works

- A more realistic case of a Neutron star with Quark core in $f(R, T)$ gravity would be relevant for the recent observations on binary millisecond pulsar⁸.
- $f(R)$ based theory of MOND has system dependent length scales such as $r_g = \frac{GM}{c^2}$ and $\ell_M = \sqrt{\frac{GM}{a_0}}$ which are introduced at the level of action, $S = -\frac{c^3}{16\pi G r_g \ell_M} \int f(R) \sqrt{-g} d^4x$. This posed a riddle which we would like to examine in the framework of $f(R, T)$ gravity.
- We would also like to see whether $f(R, T)$ can actually explain the existence of super-Chandrasekar white dwarfs.

⁸J. Antoniadis et al. Science 340 1233232 (2013), P. B. Demorest et al., Nature 467, 1081 (2010), E. Fonseca et al., Astrophys. J. 832, 167 (2016), H. T. Cromartie et al., Nat. Astron. (2019)

List of Publications

- A. Mathew and M. K. Nandy, General relativistic calculations for white dwarfs, Res. Astron. Astrophys. **17**(6), 061 (2017)
- A. Mathew and M. K. Nandy, Effect of minimal length uncertainty on the mass-radius relation of white dwarfs, Ann. Phys.(N. Y.) **393**, 184 (2018)
- A. Mathew and M. K. Nandy, Noncommutative dispersion relation and mass-radius relation of white dwarfs, Res. Astron. Astrophys., **18**, 145 (2018)
- Prospect of Chandrasekhar's limit against modified dispersion relation (2019) [submitted]
- Stellar structure of quark stars in a modified Starobinsky gravity (2019) [submitted]
- Reassurance of Chandrasekhars limit against quantum gravitational fluctuations (2019) [preparing]
- GUP Relativistic White Dwarfs (2019) [preparing]
- Non-commutative Relativistic White Dwarfs (2019) [preparing]

THANK YOU